Package ‘manMetaVAR’

October 26, 2025

Title Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients

Version 0.9.3

Description Research compendium for the manuscript
Pesigan, I. J. A., et al. (In Preparation).
Multivariate Meta-Analysis of Vector Autoregressive Model Coefficients.
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URL `https://github.com/jeksterslab/manMetaVAR`, `https://jeksterslab.github.io/manMetaVAR/`, `https://osf.io/uenr7/`, `https://doi.org/10.0000/0000000000`

BugReports `https://github.com/jeksterslab/manMetaVAR/issues`

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Encoding UTF-8

LazyData true

Roxygen list(markdown = TRUE)

Depends R (>= 4.0.0), OpenMx, fitDTVARMxID, metaVAR

Imports simStateSpace, mlVAR

Remotes jeksterslab/fitDTVARMxID, jeksterslab/metaVAR

Suggests knitr, rmarkdown, testthat

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NeedsCompilation no

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`coef.manmetavar.dtvar` *Parameter Estimates (FitDTVAR)*

Description

Parameter Estimates (FitDTVAR)

Usage

```
## S3 method for class 'manmetavar.dtvar'
coef(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
```

```
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)
```

Arguments

object	Object of class manmetavar.dtvar.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
...	additional arguments.

Value

Returns a list of vectors of parameter estimates.

Author(s)

Ivan Jacob Agaloos Pesigan

`coef.manmetavar.metavar` *Parameter Estimates (FitMetaVAR)*

Description

Parameter Estimates (FitMetaVAR)

Usage

```
## S3 method for class 'manmetavar.metavar'
coef(object, ...)
```

Arguments

`object`	Object of class `manmetavar.metavar`.
`...`	additional arguments.

Value

Returns a vector of estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

`Compress` *Compress Replication*

Description

Compress Replication

Usage

```
Compress(taskid, repid, output_folder)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`Dynamics` *Process Dynamics*

Description

Process Dynamics

Usage

```
Dynamics(dynamics)
```

Arguments

`dynamics`	1, 2, or 3. 1 for stable reciprocal regulation, 2 for escalating co-activation, and 3 for adaptive recovery.

See Also

Other Data Generation Functions: `GenData()`

Examples

```
## Not run:
# stable reciprocal regulation,
Dynamics(dynamics = 1)
# escalating co-activation
Dynamics(dynamics = 2)
# adaptive recovery
Dynamics(dynamics = 3)

## End(Not run)
```

`FitDTVAR` *Fit the Model using the fitDTVARMxID Package*

Description

The function fits the model using the fitDTVARMxID::fitDTVARMxID package.

Usage

```
FitDTVAR(data, ncores = NULL, seed = NULL)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`ncores`	Positive integer. Number of cores to use.
`seed`	Integer. Random seed.

See Also

Other Model Fitting Functions: `FitMLVAR()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
print(fit)
summary(fit)
coef(fit)
vcov(fit)

## End(Not run)
```

`FitMetaVAR` *Multivariate Meta-Analysis using the metaVAR Package*

Description

The function performs multivariate meta-snalysis using the metaVAR::metaVAR package.

Usage

```
FitMetaVAR(fit, ncores = NULL)
```

Arguments

`fit`	R object. Output of the `FitDTVAR()` function.
`ncores`	Positive integer. Number of cores to use.

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitDTVAR(
  data = data,
  ncores = parallel::detectCores()
)
pooled <- FitMetaVAR(
  fit = fit,
  ncores = parallel::detectCores()
)
summary(pooled)
print(pooled)
coef(pooled)
vcov(pooled)

## End(Not run)
```

`FitMLVAR` *Fit the Model using the mlVAR Package*

Description

The function fits the model using the mlVAR::mlVAR package.

Usage

```
FitMLVAR(data, ncores = NULL)
```

Arguments

`data`	R object. Output of the `GenData()` function.
`ncores`	Positive integer. Number of cores to use.

See Also

Other Model Fitting Functions: `FitDTVAR()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
fit <- FitMLVAR(data = data)
print(fit)
summary(fit)
```

```
## End(Not run)
```

GenData *Simulate Data*

Description

The function simulates data using the `simStateSpace::SimSSMIVary()` function.

Usage

```
GenData(taskid)
```

Arguments

`taskid`	Positive integer. Task ID.

See Also

Other Data Generation Functions: `Dynamics()`

Examples

```
## Not run:
set.seed(42)
data <- GenData(taskid = 1)
print(data)
summary(data)
plot(data)

## End(Not run)
```

---

params *Simulation Parameters*

Description

Simulation Parameters

Usage

```
params
```

Format

A dataframe with 25 rows and 3 columns:

taskid Simulation Task ID.

n Sample size.

time Number of measurement occassions.

dynamics Process dynamics. `1` for stable reciprocal regulation, `2` for escalating co-activation, and `3` for adaptive recovery.

Author(s)

Ivan Jacob Agaloos Pesigan

`plot.manmetavar.data` *Plot Method for an Object of Class* `manmetavar.data`

Description

Plot Method for an Object of Class `manmetavar.data`

Usage

```
## S3 method for class 'manmetavar.data'
plot(x, ...)
```

Arguments

`x`	Object of class `manmetavar.data`.
`...`	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

print.manmetavar.data *Print Method for an Object of Class* manmetavar.data

Description

Print Method for an Object of Class manmetavar.data

Usage

```
## S3 method for class 'manmetavar.data'
print(x, ...)
```

Arguments

x	Object of class manmetavar.data.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

print.manmetavar.dtvar *Print Method (FitDTVAR)*

Description

Print Method (FitDTVAR)

Usage

```
## S3 method for class 'manmetavar.dtvar'
print(
  x,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
```

```
  digits = 4,
  ...
)
```

Arguments

`x`	Object of class `manmetavar.dtvar`.
`means`	Logical. If `means = TRUE`, return means. Otherwise, the function returns raw estimates.
`alpha`	Logical. If `alpha = TRUE`, include estimates of the `alpha` vector, if available. If `alpha = FALSE`, exclude estimates of the `alpha` vector.
`beta`	Logical. If `beta = TRUE`, include estimates of the `beta` matrix, if available. If `beta = FALSE`, exclude estimates of the `beta` matrix.
`nu`	Logical. If `nu = TRUE`, include estimates of the `nu` vector, if available. If `nu = FALSE`, exclude estimates of the `nu` vector.
`psi`	Logical. If `psi = TRUE`, include estimates of the `psi` matrix, if available. If `psi = FALSE`, exclude estimates of the `psi` matrix.
`theta`	Logical. If `theta = TRUE`, include estimates of the `theta` matrix, if available. If `theta = FALSE`, exclude estimates of the `theta` matrix.
`converged`	Logical. Only include converged cases.
`grad_tol`	Numeric scalar. Tolerance for the maximum absolute gradient if `converged = TRUE`.
`hess_tol`	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if `converged = TRUE`.
`vanishing_theta`	Logical. Test for measurement error variance going to zero if `converged = TRUE`.
`theta_tol`	Numeric. Tolerance for vanishing theta test if `converged` and `theta_tol` are `TRUE`.
`digits`	Integer indicating the number of decimal places to display.
`...`	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

`print.manmetavar.metavar` *Print Method (FitMetaVAR)*

Description

Print Method (FitMetaVAR)

Usage

```
## S3 method for class 'manmetavar.metavar'
print(x, alpha = 0.05, digits = 4, ...)
```

Arguments

x	Object of class manmetavar.metavar.
alpha	Numeric vector. Significance level α.
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

---

print.manmetavar.mlvar

Print Method (FitMLVAR)

Description

Print Method (FitMLVAR)

Usage

```
## S3 method for class 'manmetavar.mlvar'
print(
  x,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

Arguments

x	Object of class manmetavar.mlvar.
show	Tables to show.
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

`Sim` *Simulation Replication*

Description

Simulation Replication

Usage

```
Sim(taskid, repid, output_folder, overwrite, integrity, seed)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.
`seed`	Integer. Random seed.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitDTVAR` *Simulation Replication - FitDTVAR*

Description

Simulation Replication - FitDTVAR

Usage

```
SimFitDTVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the Sim function.

Value

The output is saved as an external file in output_folder.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFitMetaVAR *Simulation Replication - FitMetaVAR*

Description

Simulation Replication - FitMetaVAR

Usage

```
SimFitMetaVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimFitMLVAR` *Simulation Replication - FitMLVAR*

Description

Simulation Replication - FitMLVAR

Usage

```
SimFitMLVAR(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

`taskid`	Positive integer. Task ID.
`repid`	Positive integer. Replication ID.
`output_folder`	Character string. Output folder.
`seed`	Integer. Random seed.
`suffix`	Character string. Output of `manCTMed:::.SimSuffix()`.
`overwrite`	Logical. Overwrite existing output in `output_folder`.
`integrity`	Logical. If `integrity = TRUE`, check for the output file integrity when `overwrite = FALSE`.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

SimFN *Simulation File Name*

Description

Simulation File Name

Usage

```
SimFN(output_type, output_folder, suffix)
```

Arguments

output_type	Character string. Output type. Valid values include "data", "fit-dt-var-mx", "fit-meta-var-mx", and "fit-ml-var".
output_folder	Character string. Output folder.
suffix	Character string. Output of manCTMed:::.SimSuffix().

Value

Returns a character string file name with the output_folder in the OS-specific format.

SimGenData *Simulation Replication - GenData*

Description

Simulation Replication - GenData

Usage

```
SimGenData(taskid, repid, output_folder, seed, suffix, overwrite, integrity)
```

Arguments

taskid	Positive integer. Task ID.
repid	Positive integer. Replication ID.
output_folder	Character string. Output folder.
seed	Integer. Random seed.
suffix	Character string. Output of manCTMed:::.SimSuffix().
overwrite	Logical. Overwrite existing output in output_folder.
integrity	Logical. If integrity = TRUE, check for the output file integrity when overwrite = FALSE.

Details

This function is executed via the `Sim` function.

Value

The output is saved as an external file in `output_folder`.

Author(s)

Ivan Jacob Agaloos Pesigan

`SimProj` *Simulation Project Name*

Description

Simulation Project Name

Usage

```
SimProj()
```

Value

Returns the project name as a character string.

Author(s)

Ivan Jacob Agaloos Pesigan

`summary.manmetavar.data` *Summary Method for an Object of Class* `manmetavar.data`

Description

Summary Method for an Object of Class `manmetavar.data`

Usage

```
## S3 method for class 'manmetavar.data'
summary(object, ...)
```

Arguments

`object`	Object of class `manmetavar.data`.
`...`	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

---

summary.manmetavar.dtvar *Summary Method (FitDTVAR)*

Description

Summary Method (FitDTVAR)

Usage

```
## S3 method for class 'manmetavar.dtvar'
summary(
  object,
  means = FALSE,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  digits = 4,
  ...
)
```

Arguments

object	Object of class `manmetavar.dtvar`.
means	Logical. If `means = TRUE`, return means. Otherwise, the function returns raw estimates.
alpha	Logical. If `alpha = TRUE`, include estimates of the `alpha` vector, if available. If `alpha = FALSE`, exclude estimates of the `alpha` vector.
beta	Logical. If `beta = TRUE`, include estimates of the `beta` matrix, if available. If `beta = FALSE`, exclude estimates of the `beta` matrix.
nu	Logical. If `nu = TRUE`, include estimates of the `nu` vector, if available. If `nu = FALSE`, exclude estimates of the `nu` vector.
psi	Logical. If `psi = TRUE`, include estimates of the `psi` matrix, if available. If `psi = FALSE`, exclude estimates of the `psi` matrix.

`theta`	Logical. If `theta = TRUE`, include estimates of the `theta` matrix, if available. If `theta = FALSE`, exclude estimates of the `theta` matrix.
`converged`	Logical. Only include converged cases.
`grad_tol`	Numeric scalar. Tolerance for the maximum absolute gradient if `converged = TRUE`.
`hess_tol`	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if `converged = TRUE`.
`vanishing_theta`	Logical. Test for measurement error variance going to zero if `converged = TRUE`.
`theta_tol`	Numeric. Tolerance for vanishing theta test if `converged` and `theta_tol` are `TRUE`.
`digits`	Integer indicating the number of decimal places to display.
`...`	additional arguments.

Author(s)

Ivan Jacob Agaloos Pesigan

`summary.manmetavar.metavar` *Summary Method (FitMetaVAR)*

Description

Summary Method (FitMetaVAR)

Usage

```
## S3 method for class 'manmetavar.metavar'
summary(object, alpha = 0.05, digits = 4, ...)
```

Arguments

`object`	Object of class `manmetavar.metavar`.
`alpha`	Numeric vector. Significance level α.
`digits`	Integer indicating the number of decimal places to display.
`...`	additional arguments.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

summary.manmetavar.mlvar *Summary Method (FitMLVAR)*

Description

Summary Method (FitMLVAR)

Usage

```
## S3 method for class 'manmetavar.mlvar'
summary(
  object,
  show = c("fit", "temporal", "contemporaneous", "between"),
  digits = 4,
  ...
)
```

Arguments

object	Object of class manmetavar.mlvar.
show	Tables to show.
digits	Integer indicating the number of decimal places to display.
...	additional arguments.

Value

Returns a matrix of estimates, standard errors, test statistics, degrees of freedom, p-values, and confidence intervals.

Author(s)

Ivan Jacob Agaloos Pesigan

vcov.manmetavar.dtvar *Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)*

Description

Sampling Covariance Matrix of the Parameter Estimates (FitDTVAR)

Usage

```
## S3 method for class 'manmetavar.dtvar'
vcov(
  object,
  alpha = TRUE,
  beta = TRUE,
  nu = TRUE,
  psi = TRUE,
  theta = TRUE,
  converged = TRUE,
  grad_tol = 0.01,
  hess_tol = 1e-08,
  vanishing_theta = TRUE,
  theta_tol = 0.001,
  ...
)
```

Arguments

object	Object of class manmetavar.dtvar.
alpha	Logical. If alpha = TRUE, include estimates of the alpha vector, if available. If alpha = FALSE, exclude estimates of the alpha vector.
beta	Logical. If beta = TRUE, include estimates of the beta matrix, if available. If beta = FALSE, exclude estimates of the beta matrix.
nu	Logical. If nu = TRUE, include estimates of the nu vector, if available. If nu = FALSE, exclude estimates of the nu vector.
psi	Logical. If psi = TRUE, include estimates of the psi matrix, if available. If psi = FALSE, exclude estimates of the psi matrix.
theta	Logical. If theta = TRUE, include estimates of the theta matrix, if available. If theta = FALSE, exclude estimates of the theta matrix.
converged	Logical. Only include converged cases.
grad_tol	Numeric scalar. Tolerance for the maximum absolute gradient if converged = TRUE.
hess_tol	Numeric scalar. Tolerance for Hessian eigenvalues; eigenvalues must be strictly greater than this value if converged = TRUE.
vanishing_theta	Logical. Test for measurement error variance going to zero if converged = TRUE.
theta_tol	Numeric. Tolerance for vanishing theta test if converged and theta_tol are TRUE.
...	additional arguments.

Value

Returns a list of sampling variance-covariance matrices.

Author(s)

Ivan Jacob Agaloos Pesigan

---

vcov.manmetavar.metavar

Sampling Covariance Matrix of the Parameter Estimates (FitMetaVAR)

Description

Sampling Covariance Matrix of the Parameter Estimates (FitMetaVAR)

Usage

```
## S3 method for class 'manmetavar.metavar'
vcov(object, ...)
```

Arguments

`object`	Object of class `manmetavar.metavar`.
`...`	additional arguments.

Value

Returns the sampling variance-covariance matrix of the estimated parameters.

Author(s)

Ivan Jacob Agaloos Pesigan

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